

LEC1

◆ Software Engineering

1.

Software engineering mainly focuses on:

- A) Writing code quickly
 - B) Managing hardware systems
 - C) Developing and maintaining software systems
 - D) Testing only
-

2.

Which attribute belongs to the transitional category?

- A) Security
 - B) Efficiency
 - C) Portability
 - D) Reliability
-

3.

Maintenance attributes include:

- A) Reusability
 - B) Usability
 - C) Scalability
 - D) Security
-

4.

Operational attributes are concerned with:

- A) Platform migration
 - B) System performance and usability
 - C) Code structure
 - D) System redesign
-

◆ DevOps

5.

DevOps is best described as:

- A) A programming language
 - B) A testing technique
 - C) A combination of development and operations
 - D) A database system
-

6.

The main goal of DevOps is:

- A) Increase system complexity
 - B) Delay deployment
 - C) Improve delivery speed and quality
 - D) Reduce coding
-

◆ **Evolution vs Maintenance**

7.

Software maintenance refers to:

- A) Continuous improvement
 - B) System redesign
 - C) Preserving software functionality
 - D) Replacing systems
-

8.

Software evolution refers to:

- A) Bug fixing only
 - B) System documentation
 - C) Gradual system improvement
 - D) Code deletion
-

9.

Maintenance activities are typically:

- A) Unstructured
- B) Planned
- C) Random
- D) Optional

10.

Evolution mainly deals with:

- A) Testing
 - B) System replacement
 - C) Changes over time
 - D) Documentation
-

◆ **Software Evolution**

11.

Lehman is معروف بـ:

- A) DevOps
 - B) Software testing
 - C) Evolution laws
 - D) UML
-

12.

The law of continuing change states that:

- A) Systems remain stable
 - B) Systems must be updated continuously
 - C) Systems do not change
 - D) Systems are replaced
-

13.

Increasing complexity means:

- A) Systems simplify over time
 - B) Complexity remains constant
 - C) Complexity increases unless managed
 - D) Systems become static
-

14.

Continuing growth refers to:

- A) Decreasing features
- B) Stable functionality

- C) Increasing functionality
 - D) Removing modules
-

15.

Declining quality occurs when:

- A) System is updated
 - B) System is maintained
 - C) System is not adapted
 - D) System is tested
-

◆ **FOSS**

16.

FOSS stands for:

- A) Free Online Software System
 - B) Free Open System Software
 - C) Free and Open Source Software
 - D) Flexible Operating Software System
-

17.

The term “free” in FOSS refers to:

- A) Zero cost only
 - B) License expiration
 - C) Freedom of use and modification
 - D) Limited access
-

18.

Open source emphasizes:

- A) Software price
 - B) Code accessibility
 - C) License cost
 - D) Hardware compatibility
-

19.

FOSS generally has:

- A) Strict restrictions
 - B) Moderate restrictions
 - C) Relaxed or no restrictions
 - D) Full ownership limits
-

◆ **Maintenance**

20.

Software maintenance is needed because:

- A) Systems are perfect
 - B) Errors may still exist
 - C) Systems are temporary
 - D) Users are limited
-

21.

Swanson classified maintenance into:

- A) Two types
 - B) Three types
 - C) Four types
 - D) Five types
-

22.

Which is NOT a basic maintenance type?

- A) Corrective
 - B) Adaptive
 - C) Perfective
 - D) Predictive
-

23.

Preventive maintenance was introduced by:

- A) IEEE
- B) ISO
- C) Swanson
- D) Lehman

24.

Corrective maintenance focuses on:

- A) Adding features
 - B) Fixing defects
 - C) Improving design
 - D) Updating system
-

25.

Adaptive maintenance is related to:

- A) Bug fixing
 - B) Performance
 - C) Environment changes
 - D) Documentation
-

◆ **Models & Processes**

26.

SMLC stands for:

- A) Software Management Life Cycle
 - B) Software Maintenance Life Cycle
 - C) System Modeling Life Cycle
 - D) Software Modern Life Cycle
-

27.

IEEE 1219 defines:

- A) 5 phases
 - B) 6 phases
 - C) 7 phases
 - D) 8 phases
-

28.

SCM is responsible for:

- A) Coding
- B) Testing

- C) Managing changes
 - D) Designing
-

29.

SCM ensures:

- A) Fast coding
 - B) Controlled modifications
 - C) Less testing
 - D) No updates
-

◆ **Reengineering**

30.

Reengineering is:

- A) Creating new systems
 - B) Testing systems
 - C) Modifying existing systems
 - D) Deleting systems
-

31.

Reengineering includes:

- A) Testing only
 - B) Reverse only
 - C) Reverse and forward engineering
 - D) Design only
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32.

Reverse engineering is:

- A) Code development
 - B) System modification
 - C) System understanding
 - D) System deletion
-

33.

One advantage of reengineering:

- A) Increased cost
 - B) Increased risk
 - C) Reduced cost
 - D) Delayed delivery
-

34.

A disadvantage of reengineering:

- A) Easy implementation
 - B) High flexibility
 - C) Limited improvement
 - D) Better maintainability
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🟢 **Answer Key** (متبصش غیر بعد ما تحل) 😊

- 1-C
- 2-C
- 3-C
- 4-B
- 5-C
- 6-C
- 7-C
- 8-C
- 9-B
- 10-C
- 11-C
- 12-B
- 13-C
- 14-C
- 15-C
- 16-C
- 17-C
- 18-B
- 19-C
- 20-B
- 21-B
- 22-D
- 23-B
- 24-B

25-C

26-B

27-C

28-C

29-B

30-C

31-C

32-C

33-C

34-C
